

Murisphere Workflow Tutorial

Technician + Facility Manager Workflow Guide

Murisphere Operations

2026-03-05

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Purpose

This tutorial is organized by **typical vivarium workflows** so teams can execute daily operations quickly and consistently.

Roles

- Technician
- PI / Facility Manager
- Admin

1. Quick Start

1. Log in from phone/tablet/browser.
2. Set **Scan Base** URL to a phone-reachable host.
3. Print cards at 100% scale.
4. Validate one real scan before room rounds.

2. Mouse Technician Daily Workflows

2.1 Start-of-shift room prep

1. Review active alerts and prioritize high severity cages.
2. Review assigned tasks and due windows.
3. Confirm protocol blocks are addressed.

2.2 Phone scan of printed cage card

1. Scan printed QR with phone camera (not the 1D barcode strip).
2. Open link in browser.
3. Verify cage ID/location.
4. Update counts/status/notes.
5. Save and confirm persistence.

2.3 Census and reconciliation

1. Start census session.
2. Scan each cage and record observed counts.
3. Complete session and resolve mismatches.

2.4 Breeding operations

1. Create/manage breeding pairs.
2. Log timed mating, plug check, weaning, transfer events.
3. Review pair productivity and retire non-productive pairs.

2.5 Health and welfare

1. Run health rounds and log findings.
2. Open vet cases and treatments.
3. Record mortality + necropsy requirement.
4. Record euthanasia events.

2.6 Tagging and samples

1. Add animal tags (ear tag/chip/tube/well).
2. Create sample records.
3. Advance sample status (collected→shipped→received→resulted).
4. Verify sample chain-of-custody events.

3. Facility Manager Daily Workflows

3.1 Morning operations review

1. Review alert feed and severity distribution.
2. Acknowledge/escalate unresolved alerts.
3. Dispatch alert notifications.

3.2 Capacity and room operations

1. Review capacity/quotas and cage-space forecasts.
2. Review consolidation opportunities.
3. Prioritize cage wash and transfer workflows.

3.3 Recommendation lifecycle

1. Generate recommendations.
2. Decision each recommendation: accept, adjust, ignore, complete.
3. Track outcomes by recommendation type/status.

3.4 Planner scenarios

1. Create scenario with demand and constraints.
2. Attach project-level demand.
3. Evaluate projected deficit and risk.
4. Review plan snapshots and assign corrective actions.

4. Regulatory and Compliance Workflows

1. Protocol expiry monitoring and edit hard-stop.
2. Deviation/CAPA logging and closure.
3. Quarantine intake/clearance tracking.
4. Qualification expiration monitoring.
5. E-signature and attachment evidence.
6. Full audit review.

5. Research Support Workflows

1. Pedigree analysis for breeding decisions.
2. Genotyping order submission.
3. Provider callback result ingestion.
4. Mendelian/deviation monitoring.
5. Reporting/export/integration jobs.

6. Visual Walkthrough

6.1 Login and Role-aware Access

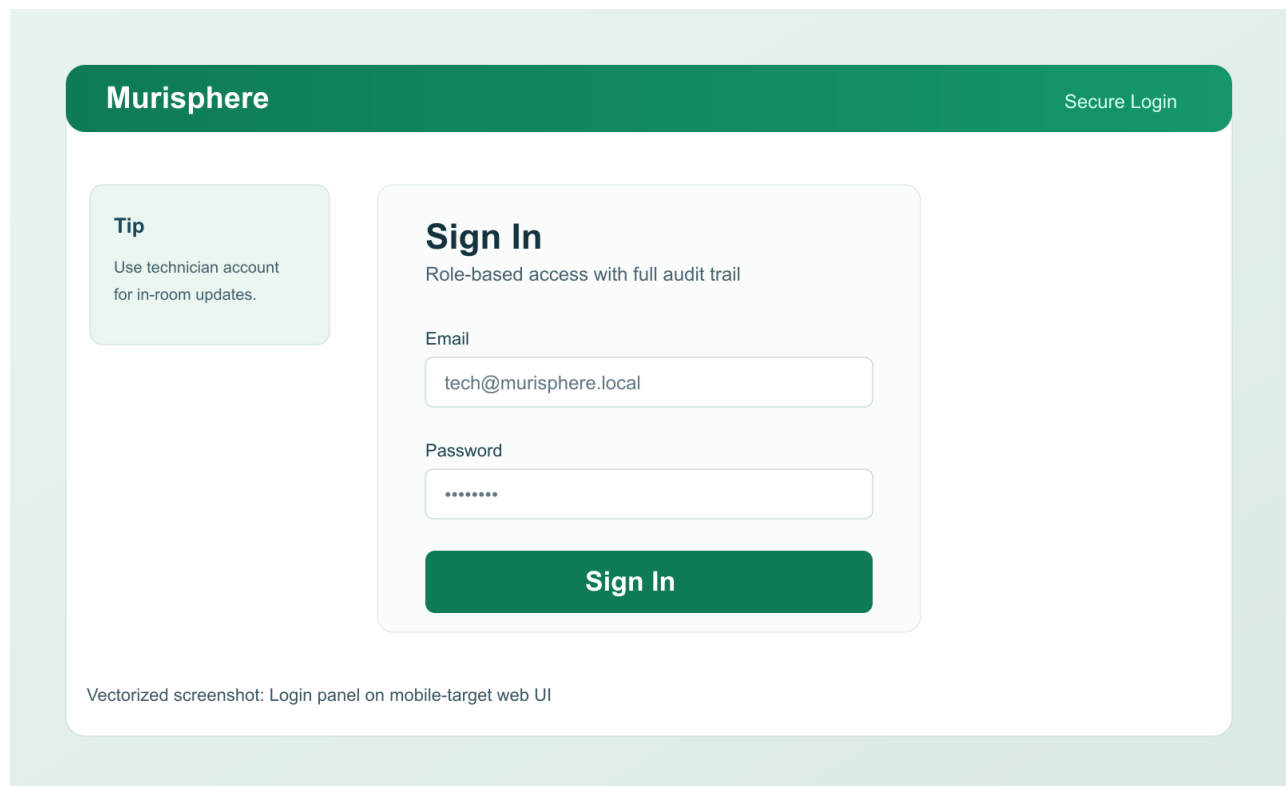


Figure 1: Login screen with role-aware access controls

Murisphere Cage Card (Rendered Preview)

Designed for fast room-floor scanning, complete husbandry context, and compliance readiness

Cage: F1-L03-C0215
Group Owner: Dr. Meosha Hudson
Group Name: Hudson Lab
Projects: L03-PRJ-01, L03-PRJ-05
Location: Room 1, Breeding Rack 1, Slot 6A

Protocol: IACUC-2026-0312
Description: PdxCre longitudinal cohort
Breeding: Pairing 35 (Timed Mating)
Protocol Expires: 12/31/2026
Population: M2 / F3 / T5


Scan with phone camera

ID	Sex	DOB	Genotype	Status
258	M	10/10/2024	PdxCre Tg/Azelia	Breeder
1L	F	10/28/2024	PdxCre +/-tg	Breeder
1L1R	F	10/28/2024	PdxCre +/-+	Breeder

Litters

#	DOB	Born	Survived	M/F
1	11/12/2024	3	3	2/1

#	DOB	Born	Survived	M/F
2	01/08/2025	5	5	1/4

Figure 2: Complete cage card with owner, protocol, animal table, litter table, QR and barcode

Scan/Edit Workflow

1. Enter QR token or scan link

Open Cage

C-A1-0243 (lookup 132ms)
 C57BL/6J | WT/WT | Breeding
 Room A1 / Rack R3
 Male Count Female Count

Save Changes

2. Phone Camera Scan

Open in Browser

Tap banner to open
 /scan/tok_a1_0243...

Vectorized screenshot: Scan token lookup and edit confirmation

Figure 3: Phone scan flow from printed QR to cage browser view

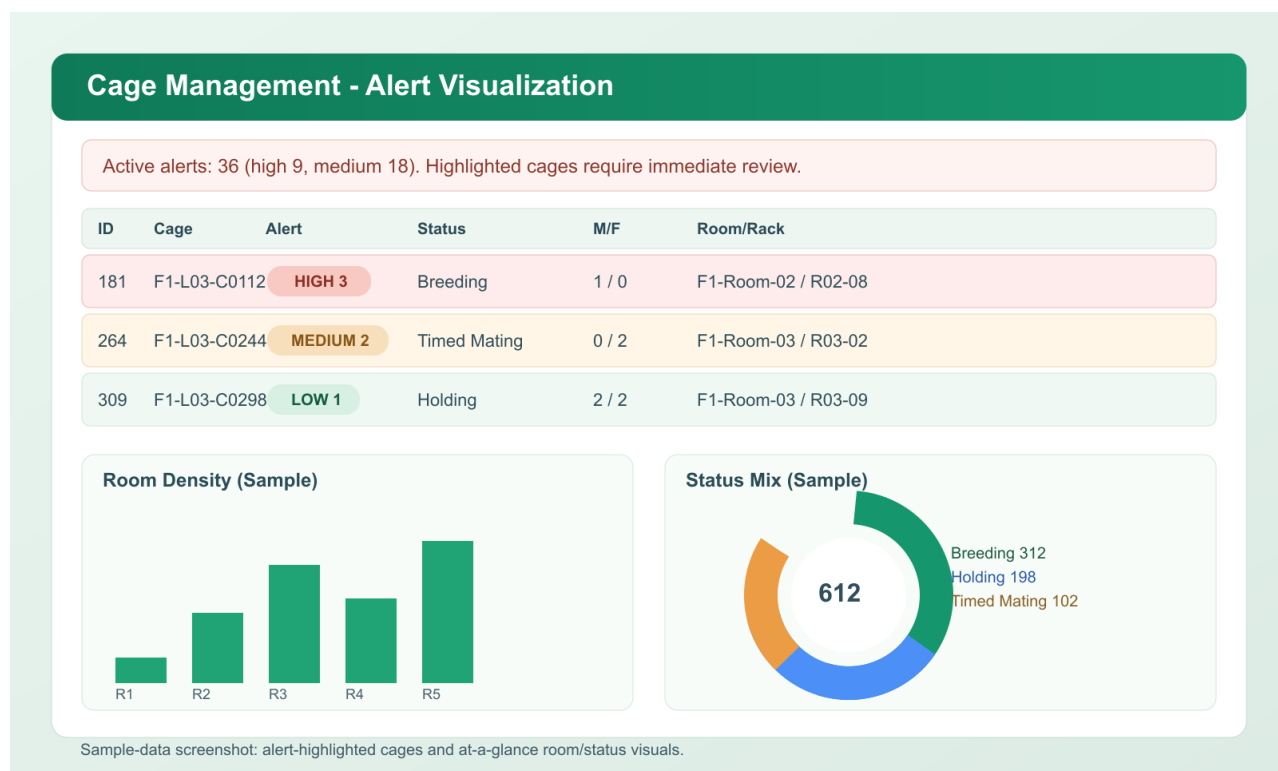


Figure 4: Cage alert highlighting and room density visualization

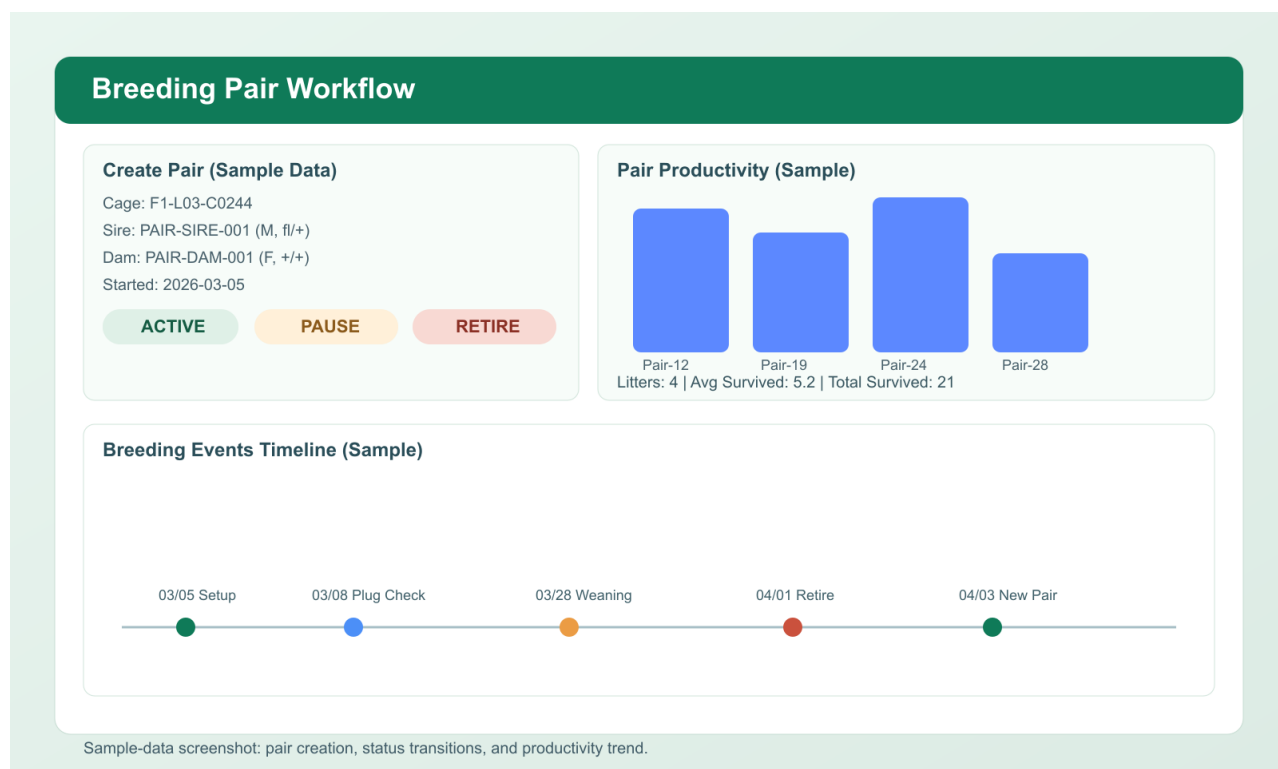


Figure 5: Breeding pair management and productivity tracking

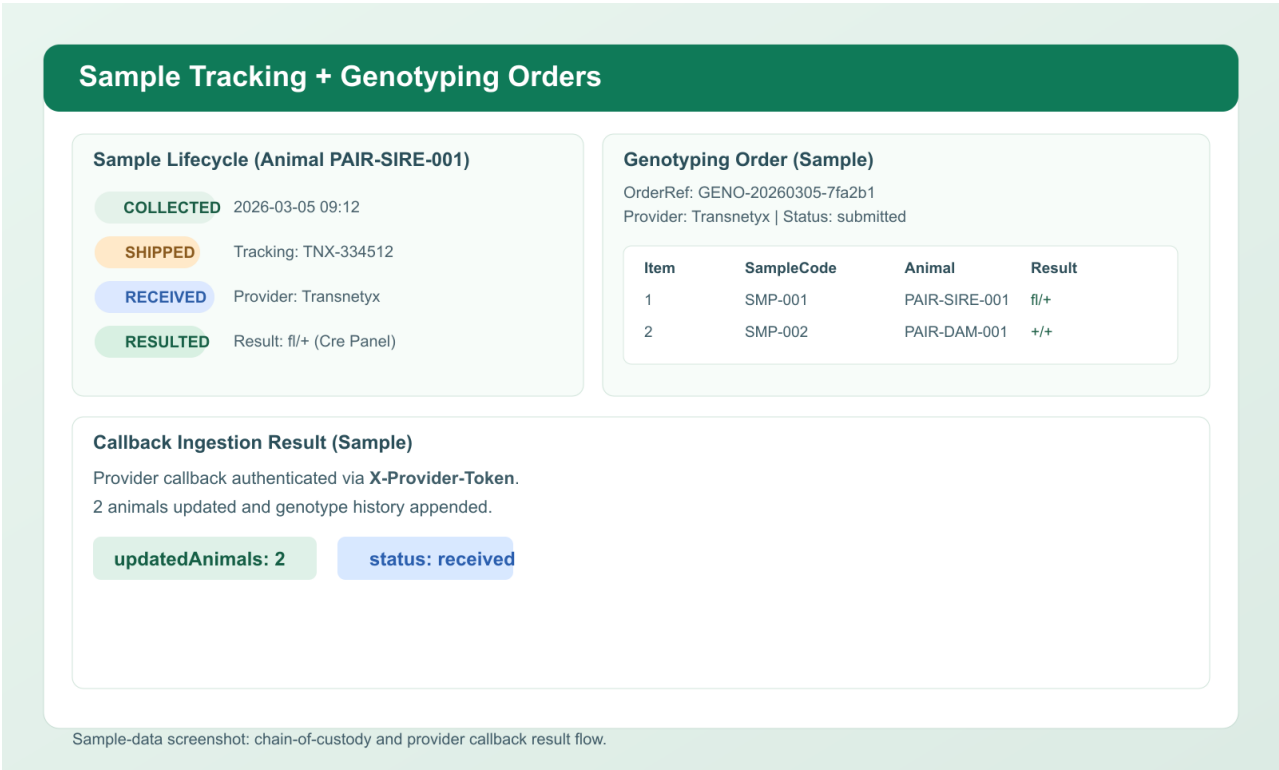


Figure 6: Sample chain-of-custody and genotyping result workflow

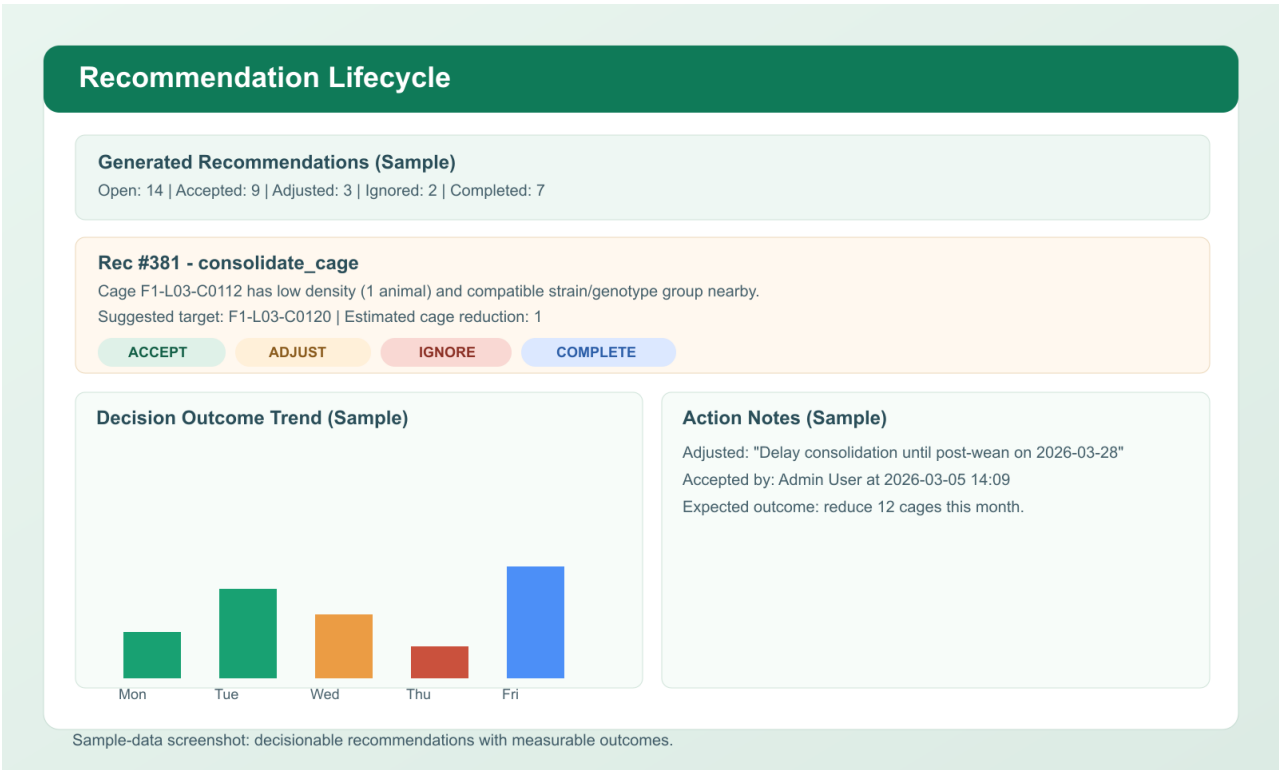


Figure 7: Recommendation lifecycle and decision outcomes

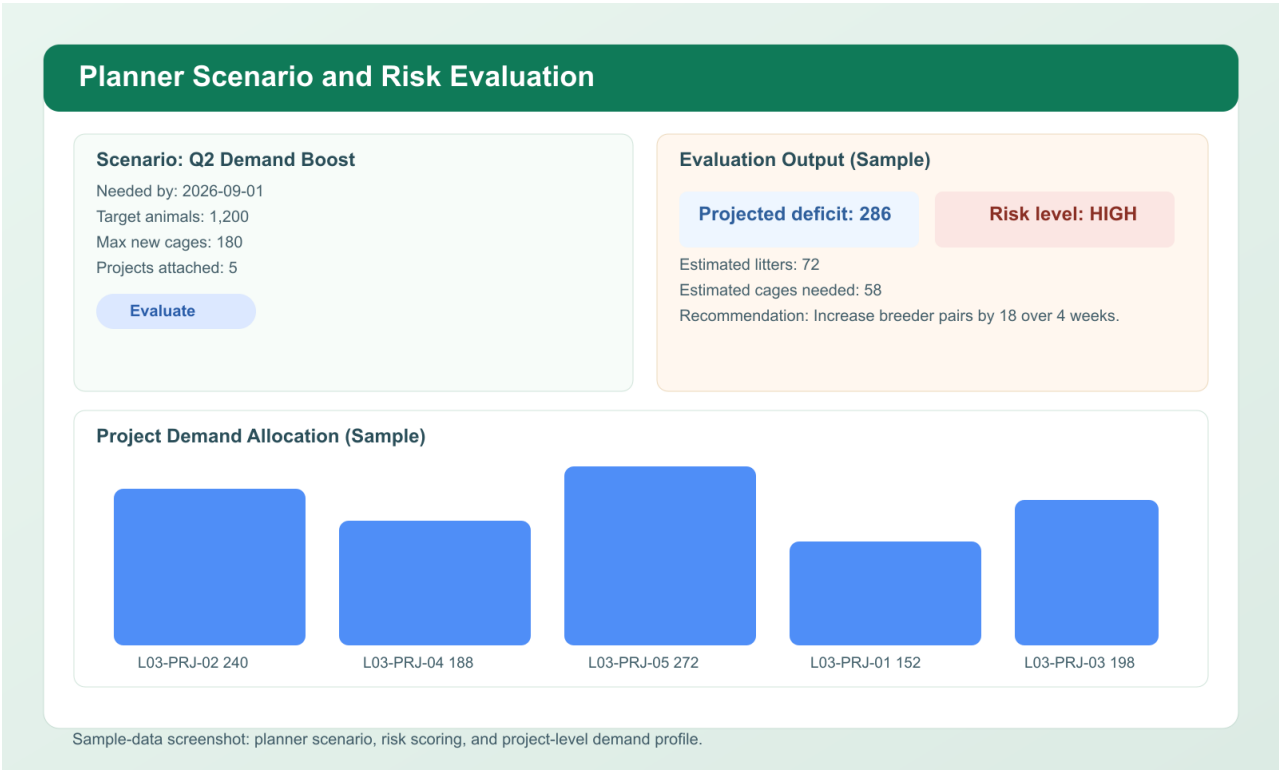


Figure 8: Planner scenario risk and deficit evaluation

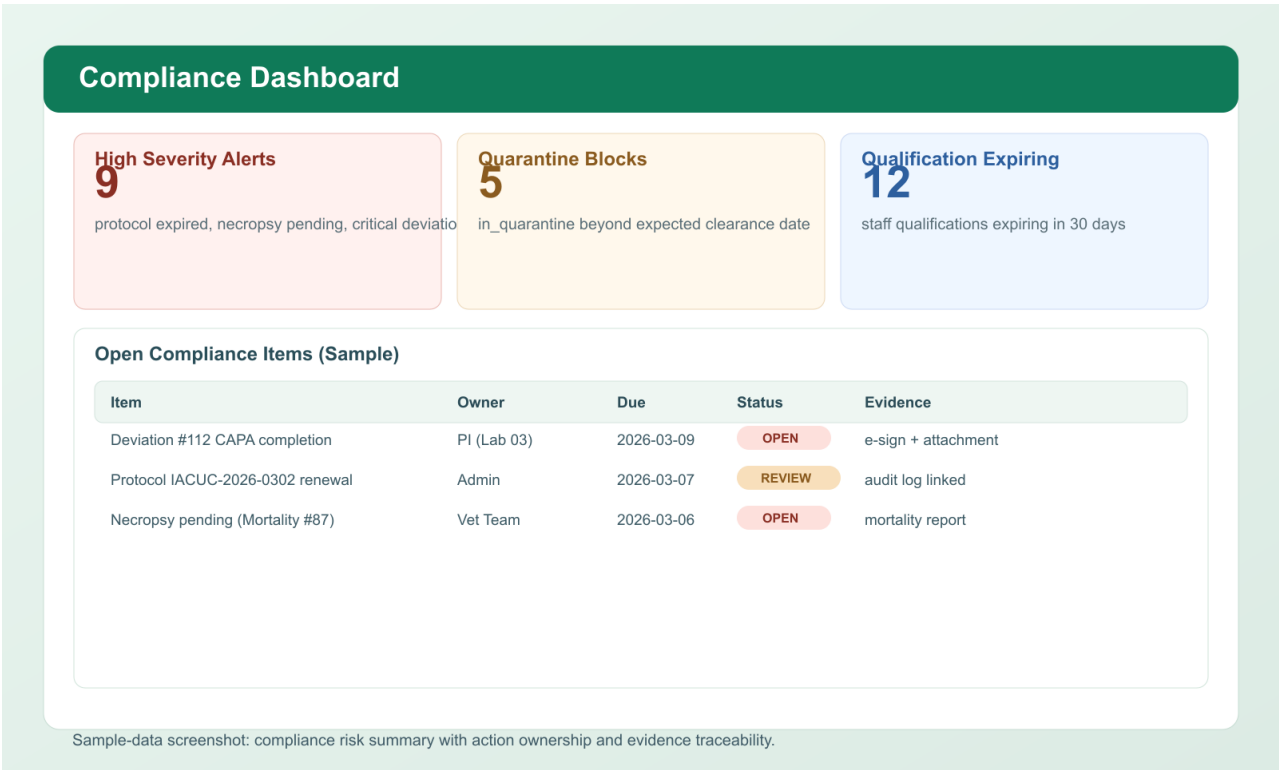


Figure 9: Compliance dashboard with alerts and open actions

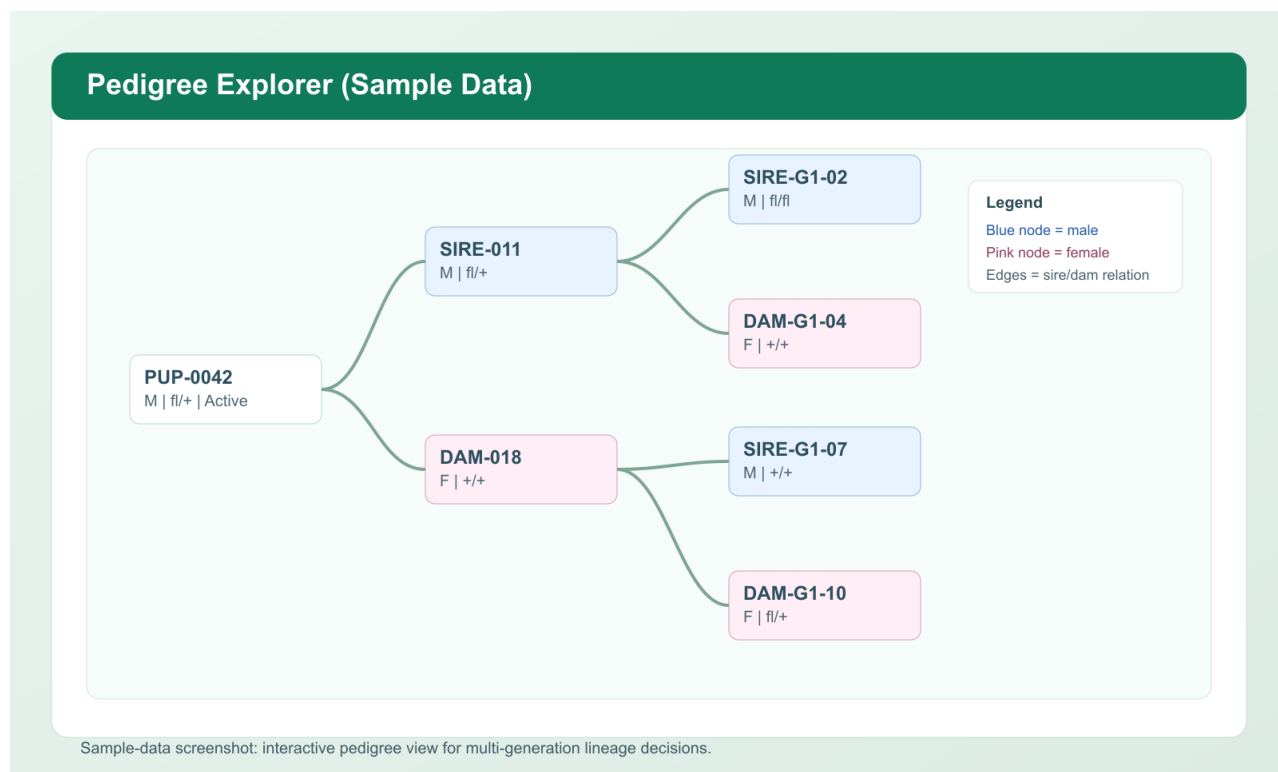


Figure 10: Multi-generation pedigree explorer

6.2 Complete Cage Card (Print + Scan)

6.3 Phone Scan and Quick Edit

6.4 Cage Alerts and Room Density

6.5 Breeding Pair Productivity

6.6 Samples and Genotyping Workflow

6.7 Recommendations and Outcomes

6.8 Planner Scenario Evaluation

6.9 Compliance Dashboard

6.10 Pedigree Explorer

7. End-of-Shift / End-of-Day Checklists

Technician

1. Confirm offline queue drained and synced.
2. Confirm health/vet/mortality/euthanasia records complete.
3. Close open room sessions.

Manager

1. Resolve or owner-assign all high severity alerts.
2. Review unresolved deviations/quarantine blocks.

3. Review planner risk and recommendation decisions.
4. Export daily summary/audit package.

8. Troubleshooting and Escalation

- No scan prompt: improve lighting and card contrast.
- Camera only sees barcode value: scan the QR square, not the CODE128 bars.
- Wrong destination: fix `Scan Base URL`; never use localhost for printed cards.
- QR broken: verify `/api/assets/qrcode.png?v=test`.
- Save queued: reconnect and verify sync before leaving room.
- Protocol hard-stop: assign valid protocol and retry.
- Genotyping callback mismatch: verify callback token and sample code mapping.